

CY 561: Natural Products Synthesis

Course Code	CY 561
Course Name	Natural Products Synthesis
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG/PG/PhD
Mutual Exclusion	NA
Approval	65th BoA (Aug 2026)

Preamble

This course introduces the principles and strategies involved in the total synthesis of structurally complex natural products. It covers the importance and classification of natural products, fundamentals of retrosynthetic analysis, synthons and synthetic equivalents, strategic bond disconnections, and the design of linear and convergent synthetic routes. Through selected case studies, students will examine major synthetic approaches such as biomimetic synthesis, macrocyclization, cascade reactions, and step-economic or protecting-group-free synthesis. Emphasis will be placed on retrosynthetic planning, chemoselectivity, regioselectivity, stereochemical control, and the evolution of synthetic strategies. Student-led literature discussions will further help develop the ability to critically analyze modern total synthesis research.

Course Modules

- **Module I: Introduction to Natural Product Synthesis:** *Importance and classification of natural products* (Importance: Natural products as sources of medicines, agrochemicals, fragrances, etc., Classification of natural products: Alkaloids, Terpenoids, Steroids, Natural pigments, etc., Physicochemical properties of secondary metabolites) (2 hours)
- **Module II: Retrosynthetic Analysis and Synthetic Strategy:** Principles of retrosynthesis, Synthons and synthetic equivalents, Functional group interconversion (FGI), Strategic bond disconnections, Linear vs. convergent synthesis, Umpolung strategies, Redox and step economy etc. (8 hours)
- **Module III: Total Synthesis of Major Classes of Natural Products:** (32 hours)
 - a) Convergent and Modular Synthesis
Representative examples: Taxol, Prostaglandins
 - b) Biomimetic Synthesis
Representative examples: Morphine, Strychnine
 - c) Macrocyclization Strategies
Representative examples: Erythromycin
 - d) Cascade and Domino Reactions in Total Synthesis
Representative examples: Reserpine
 - e) Step-Economic and Protecting-Group-Free Synthesis
Representative examples: Euphorikanin A, and Hispidospermidin
 - f) Seminal Contributions of Indian Scientists to Total Synthesis
 - g) Student-led literature presentations and discussions

Emphasis will be shown on understanding the logic and evolution of synthetic design, including: Retrosynthetic planning, Strategic bond disconnections, Chemoselectivity, regioselectivity, and Stereochemical control, Comparison of alternative synthetic routes, Evolution of synthetic strategies over time.

Text Books

1. Nicolaou K. C. *Classics in Total Synthesis of Natural Products*, Vol. I & II
2. Corey, E.J. and Cheng, X.-M., *The Logic of Chemical Synthesis*, Wiley-VCH, Weinheim. 1995

References

1. J. Clayden, N. Greeves, S. Warren and P. Wothers., (2012), *Organic Chemistry*
2. Francis A. Carey and Richard J. Sundberg, *Advanced Organic Chemistry*, Springer 2008